Mohammad Islamul Haque

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Current Positions and Affiliations

2023–present	**Postdoctoral Associate**, Yale Research Initiative on Innovation and Scale (Y-RISE), MacMillan Center, Yale University
present	**Affiliated Researcher**, Jameel Observatory CREWSnet, MIT, and Center for Natural Hazards, Climate Risk, and Earth Sustainability (CRISES), Project Sonar Bangla, MIT

Education

2019–2023	**Ph.D. in Economics**, University of Southern California Dissertation: *Essays on the Economics of Climate Change Adaptation in Developing Countries* (May 2023)
2017–2019	**M.A. in Economics**, University of Southern California
2016–2017	**M.S.S. in Economics**, University of Dhaka
2012–2015	**B.S.S. in Economics**, University of Dhaka

References

Matthew E. Kahn, Provost Professor of Economics and Spatial Sciences, Department of Economics, University of Southern California, Kaprielian Hall 300, Los Angeles, CA 90089
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Paulina Oliva, Professor of Economics and Spatial Sciences, Department of Economics, University of Southern California, Kaprielian Hall 300, Los Angeles, CA 90089
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Ahmed Mushfiq Mobarak, Jerome Kasoff '54 Professor of Management and Economics, Yale School of Management and Department of Economics, Yale University, 165 Whitney Avenue, New Haven, CT 06511
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Fields

Primary	Development Economics, Environmental Economics
Secondary	Economics of Climate Change Adaptation, Labor Mobility

Job Market Paper

Who Bears the Loss from a Flood? Prices, Rationing, and Incidence in Rural Bangladesh

Floods destroy livestock, equipment and other productive assets, and the usual presumption is that the poor bear the loss through a lower wage or rationed work. Following 5,473 rural households in Bangladesh's flood-prone delta from 2011 to 2019, matched to satellite flood maps, I ask who bears it, through which village markets, and who adapts. The presumption fails. The loss falls on capital, which the land-rich own: the largest inherited landholders lose livestock and working capital, their farms shrink, and their income from work falls by 7.6 percent of a year's income. The labor market rations work rather than repricing it; the land-poor lose nearly a quarter of their day labor at an unchanged wage, yet they adapt by turning to self-employment, and their consumption and savings show no fall. The rent falls by a fifth, but land does

not reallocate. Labor can be redeployed; capital cannot. Who bears a flood follows who owns the capital it hits, so relief aimed at the poor would miss the farms that lose capital: a grant or fairly priced index insurance to farms that lose a herd lets them rebuild within a year.

Working Papers

The Role of Land Market in Achieving the Scale for Adaptation to Climate Change: Evidence from Bangladesh

This paper studies whether climate-induced shifts in land productivity can trigger structural adaptation in agriculture by overcoming long-standing frictions in land markets. In coastal Bangladesh, rising sea levels have pushed dry-season soil salinity beyond the threshold at which conventional crops survive. Using census data on over one million farm households and a fuzzy spatial regression discontinuity at the hydrological boundary between the tidally active and mature regions of the Bengal delta, I find that salinity above 12 dS/m moves 32 to 50 decimals of land per household from crops to aquaculture and raises average farm size by 52 percent. The transition runs through the rental market rather than land sales. Reallocation is faster in villages with greater religious diversity and slower where parcels are more fragmented.

Adapting to High Heat, Failing at the Extreme: Newborn Heat Mortality over Four Decades in Bangladesh (with Anoushka Chopra)

Damage functions in climate economics typically assume that rising income and cooling technology steadily reduce the mortality cost of extreme heat. We test this with a birth-cohort decomposition for 276,031 Bangladeshi births from nine DHS rounds merged with ERA5 daily-maximum temperature. At 30–40°C, only the 1980 cohort shows a significant response; each later cohort's response is close to zero and significantly below it, and a heterogeneity-robust difference-in-differences estimator links the flattening to the staggered rollout of rural electrification, implying roughly 718,000 neonatal deaths averted. Above 40°C the effect is large for the 2010 cohort, about 43 additional deaths per 1,000 births per 10 additional days, and no test detects a decline. The tail effect implies 657 to 1,920 excess neonatal deaths per year by mid-century.

Heat and the Body: Acute Cardiometabolic Responses and the Measurement of Chronic Disease in Bangladesh

Household surveys diagnose chronic disease from a single measurement of each respondent. Using 34,879 blood-pressure readings and 33,179 fasting glucose results from three Bangladesh DHS rounds, merged with ERA5 temperature at the date and hour of measurement, I test whether they also measure the weather. A one-degree rise in the daily maximum lowers systolic blood pressure on the day it is read by 0.56 mmHg and raises next-morning fasting glucose by 1.03 mg/dL; the response is to humid heat, and a fall in haemoglobin points to dilution rather than dehydration. Measured diabetes rises 0.63 and measured hypertension falls 0.84 percentage points per degree, while reported past diagnoses do not respond, and warmer 2022 fieldwork accounts for about a third of the 7.8-point rise in measured diabetes among adults aged 35 and over since 2017–18.

From Two Daughters to a Hurried Third: Son Preference, Birth Spacing, and Mortality (with Syed Manzoor Ahmed Hanifi)

Son preference raises child mortality through sex selection, discriminatory investment and early weaning; we document a fourth channel, birth spacing. Bangladeshi mothers of two daughters conceive a third child sooner, compressing the interval below the WHO-recommended 24 months. Instrumenting a sub-24-month third conception with the sex composition of the first two births, we estimate a 10.2 percentage point higher risk of third-child neonatal death in Matlab and Chakaria surveillance data and 11.4 points in nine Bangladesh DHS rounds. The child harmed is the one the parents wanted, male with even probability. Nationally, this implies about 35,000 excess neonatal deaths over four decades.

Measuring Flood Damage to Pond Aquaculture: Satellite Evidence from Bangladesh (with Runa Laila)

We match three rounds of a Bangladeshi household panel to satellite-observed inundation to measure flood

damage to pond aquaculture. Any inundation lowers the probability of operating a pond by 2.9 percentage points, a 16 percent relative decline. Severe floods, which inundate more than half the surrounding area, reduce yield by 48 percent, revenue by 47 percent and profit by about 18,500 taka. No coping margin compensates, and household income falls by more than the pond loss alone. Such floods cost the sector about 26 million dollars a year.

Ongoing Field Experiments and Data Collection

Strategies for Scaling Up International Migration in Underutilized Routes within Asia (with Austin Davis, Cynthia Kinnan, Daliah Al-Shakhshir, and Ahmed Mushfiq Mobarak)
Randomized controlled trial on the Philippines–Japan migration corridor with Onodera User Run, a Japanese intermediary whose employer-financed model makes language and vocational training free to the migrant. Three randomizations: admission to training over roughly eleven quarterly cohorts, supplemental personal-initiative training among those admitted, and recruitment intensity across 98 barangays to identify spillovers.

OUR Japan Cohort Study (with Austin Davis, Cynthia Kinnan, and Ahmed Mushfiq Mobarak)
Five-year panel of 750 Filipino migrants already working in Japan, drawn from five consecutive deployment cohorts, tracking earnings, savings, remittances, language, family life and well-being, with modules carried from a UAE–India migrant study for corridor comparison. Led the survey design.

Adapting to Drinking Water Scarcity in Coastal Bangladesh through Community-Level Water Desalination Business (with Nahian Bin Khaled, Ahmed Mushfiq Mobarak, and Jared Stolove)
Pilot cluster-randomized trial of community-scale reverse osmosis plants under a social-enterprise model: 15 of 50 water-scarce communities receive a plant, about 100 households are surveyed per community, and the per-liter price each household pays is randomized across four levels by public lottery. Baseline October–November 2026.

Coastal Bangladesh Longitudinal Survey (with Ahmed Mushfiq Mobarak, Dev Patel, and Nahian Bin Khaled; project lead)
Household and farm panel of roughly 4,500 households in 225 villages across Khulna, Bagerhat and Satkhira, sampled along the salinity gradient with soil and water salinity measured at village and plot level. First wave September–October 2026; annual follow-ups from 2027; linked to MIT JO-CREWSnet.

Works in Progress

Effects of Increase in Decadal Monsoon Onset Variability on Farming: Evidence from Bangladesh (with Jared Stolove)

Migration to the Middle East and Women's Empowerment in Bangladesh: Evidence on Household Decision-Making and Agency (with Cecile Pierre and Frederica Mendonça)

Approaches to Studying Adaptation in Early Climate Change Battlegrounds (with Ahmed Mushfiq Mobarak and Matthew Kahn)

Underinvestment in a Profitable Adaptation Technology: The Role of Market Access

Publications and Policy Writing

Kahn, Matthew and Mohammad Islamul Haque. "Cities in the Developing World." In *Introduction to Urban Economics*, 2023. Book chapter.

Mobarak, Ahmed Mushfiq and Mohammad Islamul Haque. "Developing Scalable Pathways for International Migration: Harnessing 'Triple Wins' for Global Economic Progress." Luxembourg Institute of Socio-Economic Research (LISER), 2024. Policy paper.

Haque, Mohammad Islamul and Ahmed Mushfiq Mobarak. "Fulfilling the July Uprising's Promise: Skilled Migration for Our Youth." *The Daily Star*, 1 August 2025. Op-ed.

Conference and Seminar Presentations

2026	Allied Social Science Associations (ASSA) Annual Meeting, Philadelphia
2023	Yale Climate, Environment & Economic Growth Conference, Economic Growth Center, Yale University
2022	UC San Diego Environmental and Resource Economics Lunch Seminar
2021	Association of Environmental and Resource Economists (AERE) Summer Conference
2020	Berkeley/Sloan Summer School in Environmental and Energy Economics

Research Grants

2024	**Community Jameel**, $500,000, Co-Principal Investigator Adaptation to salinity intrusion in coastal Bangladesh
2024	**Private Enterprise Development in Low-Income Countries (PEDL)**, $60,000, and **J-PAL King Climate Action Initiative (K-CAI)**, $50,000, Co-Principal Investigator Pilot RCT of a social-enterprise reverse-osmosis water service
2022	**International Growth Centre (IGC)**, $48,000, Principal Investigator Data collection for "The Role of Land Market in Achieving the Scale for Adaptation to Climate Change"

Teaching

Instructor

Spring 2025	**ECON 489: The Economics of Climate Change Adaptation**, Yale University New undergraduate seminar; designed the syllabus, readings and assessments
2017	**Principles of Microeconomics** and **Mathematics for Economists** (Calculus I and Linear Algebra I), East West University, Dhaka Adjunct Lecturer

Teaching assistant, University of Southern California

Spring 2022	ECON 587: Urban Economics (Prof. Matthew Kahn)
Spring 2022	ECON 584: Advanced Economic Consulting and Antitrust Methods (Prof. Jeffrey Dubin)
Spring 2021	ECON 101: Principles of Microeconomics

Other Positions and Service

Spring 2025	**Lecturer**, Department of Economics, Yale University
2023–2024	**Short-Term Consultant**, The World Bank Empirical analysis of agricultural adaptation to climate shocks for *Rethinking Resilience*
Referee	International Labour Review
Memberships	American Economic Association; Association of Environmental and Resource Economists; Agricultural & Applied Economics Association

Skills, Languages and Citizenship

Methods	Randomized controlled trials, survey design and field data collection, applied microeconometrics
Software	Stata, Python, R, ArcGIS, LaTeX
Languages	English (fluent), Bangla (native) Citizenship: Bangladesh